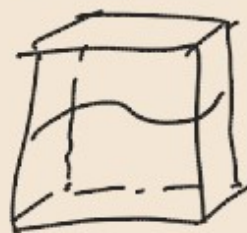


Black body radiation:

- A box of size L supports standing elec. mag. waves w/ angular freq.

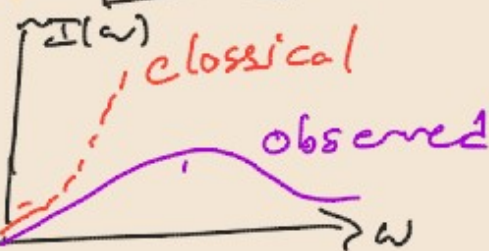
$$\omega_n = \frac{2\pi}{L} |\vec{n}| c$$



Interp.:

$$I(\omega) = \frac{1}{V} \frac{d}{d\omega} [E(\omega)] = \alpha \omega^3 k_B T$$

(classical)



* Ultraviolet Catastrophe

$\Rightarrow$ Planck Postulated:

$$E_n = \hbar \omega_n = \frac{2\pi\hbar}{L} |\vec{n}| = |\vec{p}_n| \quad \textcircled{\star}$$

Since $E_n^2 = |\vec{p}|^2 c^2 + m^2 c^4$, then these particles are massless from $\textcircled{\star}$

- Each mode of freq. ω_n can be excited an integer # j times

$$E_{\text{tot}} = j E_n = j (\hbar \omega_n)$$

- Boltzmann weight:

$$\langle E_n \rangle = \frac{\sum_{j=0}^{\infty} (j E_n) e^{-j E_n \beta}}{\sum_{j=0}^{\infty} e^{-j E_n \beta}} = \frac{\frac{d}{d\beta} \left(\sum_{j=0}^{\infty} e^{-j E_n \beta} \right)}{\sum_{j=0}^{\infty} e^{-j E_n \beta}} \quad , \quad \beta \equiv \frac{1}{k_B T}$$

$$\text{Recall } \sum_{j=0}^{\infty} x^j = \frac{1}{1-x}$$

Thus:

$$\begin{aligned} \langle E_n \rangle &= \left(-\frac{d}{d\beta} \right) \left(\frac{1}{1 - e^{-\hbar \omega_n \beta}} \right) \left[\frac{1}{(1 - e^{-\hbar \omega_n \beta})} \right] \\ &= + \left(\frac{1}{1 - e^{-\hbar \omega_n \beta}} \right)^2 (\hbar \omega_n) e^{-\hbar \omega_n \beta} / \left(\frac{1}{1 - e^{-\hbar \omega_n \beta}} \right) \\ &= \frac{\hbar \omega_n e^{-\hbar \omega_n \beta}}{1 - e^{-\hbar \omega_n \beta}} \end{aligned}$$

$$\therefore \langle E_n \rangle = \hbar \omega_n / (e^{\hbar \omega_n \beta} - 1)$$

We can take the Continuum limit, $L \rightarrow \infty$

$$E = \sum_n \langle E_n \rangle = \lim_{L \rightarrow \infty} \sum_n \langle E_n \rangle$$

$$E = \int d^3 \vec{n} \left(\frac{\hbar \omega_n}{e^{\hbar \omega_n \beta} - 1} \right)$$

Spherical coords w/ $|\vec{n}|^2 \equiv r^2$

$$\int d^3 |\vec{n}| = \int d\phi \sin(\phi) \int d\theta \int d|\vec{n}| |\vec{n}|^2$$

$$\begin{aligned} E &= \int_{-1}^1 d\cos(\phi) \int_0^{2\pi} d\theta \int_0^{\omega} d|\vec{n}| \left(\frac{|\vec{n}|^2 \hbar \omega_n}{e^{\hbar \omega_n \beta} - 1} \right) \\ &= 4\pi \int_0^{\omega} d|\vec{n}| \left(|\vec{n}|^2 \hbar \omega_n / (e^{\hbar \omega_n \beta} - 1) \right) \end{aligned}$$

but

$$\hbar \omega_n = \frac{2\pi}{L} \hbar |\vec{n}| \Leftrightarrow \frac{L^2}{4\pi^2} \omega_n^2 = |\vec{n}|^2$$

$$d|\vec{n}| = \frac{L}{2\pi} d\omega_n$$

$$E = 4\pi \hbar \left(\frac{L^2}{4\pi^2} \right) \left(\frac{L}{2\pi} \right) \int_0^{\omega L/2\pi} d\omega_n \frac{\omega_n^3}{e^{\hbar \omega_n \beta} - 1}$$

$$E = 4\pi \hbar \frac{L^3}{8\pi^3} \int_0^{\omega L/2\pi} d\omega_n \frac{\omega_n^3}{e^{\hbar \omega_n \beta} - 1}$$

$\therefore$ The intensity of light as a fⁿ of frequency is:

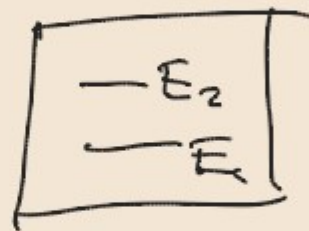
$$I(\omega) = \frac{1}{V} \frac{dE(\omega)}{d\omega} = \frac{\hbar}{\pi^2} \left(\frac{\omega^3}{e^{\hbar \omega \beta} - 1} \right)$$

Einstein Coefficients:

• Suppose we have a cavity full of atoms full of atoms w/ energy levels E_1 & E_2

Assume n_1 : E_1

n_2 : E_2



Each level is reached via a difference of $\hbar \omega$. I.e.

$$\hbar \omega = E_2 - E_1$$

A: Coeff. of spontaneous emission

B: Coeff. of Stimulated emission

$$\Leftrightarrow dn_2 = -[A + B I(\omega)] n_2$$

Since $\Gamma(\omega) = \frac{dE/d\omega}{V}$

$$\Leftrightarrow [B] \sim \frac{V}{dE/d\omega}$$

B': Coeff. of absorption $1 \rightarrow 2$

$\rightarrow$ decreases n_1 and increases n_2
by $B' I(\omega) n_1$

$$n = n_1 + n_2 \Leftrightarrow dn = 0 \Leftrightarrow dn_2 = -dn_1$$

$$dn_2 = -dn_1 = -[A + B I(\omega)] n_2 + B' I(\omega) n_1$$

• In Equilibrium $dn_2 = dn_1 = 0$ and each are determined by Boltzmann dist.

$$n_1 = N e^{-\beta E_1} \quad \text{and} \quad n_2 = N e^{-\beta E_2}$$

$$\Leftrightarrow [B' e^{-\beta E_1} - B e^{-\beta E_2}] I(\omega) = A e^{-\beta E_2}$$

$$\Gamma(\omega) = \frac{A}{(B' e^{\beta(E_2 - E_1)} - B)} \quad (\star)$$
$$= \frac{A}{B' e^{\beta \hbar \omega} - B}$$

Since we know, for a blackbody,

$$\Gamma(\omega) = \frac{\hbar}{\pi} \frac{\omega^3}{(e^{\beta \hbar \omega} - 1)}$$

and $(\star)$ must hold for any system

$$\boxed{B = B'}$$

$$\Gamma(\omega) = \frac{A}{B} \left(\frac{1}{e^{\beta \hbar \omega} - 1} \right) = \frac{\hbar}{\pi} \frac{\omega^3}{(e^{\beta \hbar \omega} - 1)}$$

$$\therefore \frac{A}{B} = \frac{\hbar \omega^3}{\pi}$$

$$\hat{N} = a^\dagger a, \quad \hat{N}|n\rangle = n|n\rangle$$

$$\hat{N}a^\dagger|n\rangle = (a^\dagger a a^\dagger)|n\rangle = a^\dagger(a^\dagger a + 1)|n\rangle$$

$$= (a^\dagger \hat{N} + a^\dagger)|n\rangle$$

$$= (a^\dagger n + a^\dagger)|n\rangle$$

$$= (n+1)a^\dagger|n\rangle$$

$$\therefore a^\dagger|n\rangle = C|n+1\rangle$$

Since $\langle n|n\rangle = 1$ and $\langle n+1|n+1\rangle = 1$

$$\langle n|a a^\dagger|n\rangle = C^2$$

$$\langle n|(\hat{N} + 1)|n\rangle = C^2 \Rightarrow C = \sqrt{n+1}$$

* Similarly we find that $a|n\rangle = \sqrt{n}|n-1\rangle$

Fermi's golden Rule: the transition rate between two states is prop. to the Matrix element squared

$$\Gamma \sim |M|^2 \delta(E_f - E_i)$$

$$M \equiv \langle f|H_{int}|i\rangle$$

$$H_{int} \sim H_I^\dagger a^\dagger + H_I a$$

* $2 \rightarrow 1$ transition: $|i\rangle \equiv |atom_2; n\omega\rangle$

$$\langle f| = \langle atom_1, n\omega+1|$$

$$M_{2-1} = \langle f|H_I^\dagger a^\dagger + H_I a|i\rangle$$

$$= \langle atom_1|H_I^\dagger|atom_2\rangle \langle n\omega+1|a^\dagger|n\omega\rangle$$

$$= M_0^\dagger \sqrt{n\omega+1}$$

$$M_0^\dagger \equiv \langle atom_1|H_I^\dagger|atom_2\rangle$$

$$|M_{2-1}|^2 = |M_0|^2 (n\omega+1)$$

$$|M_{2 \rightarrow 2}|^2 = M_0^2 \omega$$

$$\begin{aligned} \therefore dn_2 &= -dn_1 = -|M_{2 \rightarrow 2}|^2 n_2 + |M_{1 \rightarrow 2}|^2 n_1 \\ &= -|M_0|^2 (\omega + 1) n_2 + |M_0|^2 \omega n_1 \end{aligned}$$